An Explainable Evolutionary Grammar Approach for Interpretable Data Stream Classification with Concept Drift

Leandro Maciel Almeida and Leandro L. Minku, *Senior Member, IEEE*

***Abstract*—Data stream classification under concept drift presents fundamental challenges for machine learning systems, requiring models that adapt to evolving data distributions while maintaining interpretability. This paper introduces a rule-based classifier, termed EGIS, that leverages grammatical evolution to generate human-readable classification rules for non-stationary data streams. Unlike black-box ensemble methods that sacrifice transparency for accuracy, EGIS produces explicit IF-THEN rules that enable domain experts to understand, validate, and trust the model's predictions. The proposed method incorporates a multi-level self-adaptation framework that automatically adjusts evolutionary parameters in response to population diversity, performance history, and drift severity signals. A specialized mutation mechanism, termed Gene Therapy, injects discriminative patterns extracted from auxiliary decision trees directly into the evolutionary population, accelerating convergence during concept drift recovery. We introduce three transition metrics, namely Transition Change Score (TCS), Rule Instability Rate (RIR), and Average Modification Severity (AMS), that quantify rule evolution dynamics, revealing not only what the classifier predicts but how and why its decision logic changes over time. Experiments across 48 datasets encompassing abrupt, gradual, recurring, and noisy drift scenarios demonstrate that EGIS achieves competitive predictive performance (average G-Mean of 0.859 on 41 binary datasets) while providing complete interpretability. Statistical analysis using the Friedman test reveals that the performance gap between EGIS and leading ensemble methods (2–4% vs ARF and ROSE) is not statistically significant after Bonferroni correction for ARF and SRP, while EGIS outperforms ERulesD2S by over 25 percentage points. The proposed transition metrics provide quantitative insight into classifier adaptation behavior, enabling practitioners to monitor how classification rules evolve in response to changing data distributions.**

***Index Terms*—Data stream classification, concept drift, explainable AI, grammatical evolution, rule-based learning, interpretable machine learning, self-adaptation**

I. Introduction

The proliferation of real-time data sources across diverse domains, ranging from financial transaction monitoring to industrial sensor networks, has created an urgent need for classification systems capable of processing continuous data streams while adapting to evolving patterns [20]. Unlike traditional batch learning paradigms where a model is trained once on a static dataset, data stream classification must address a constellation of interrelated challenges: concept drift manifests as changes in the underlying data distribution that render previously learned models obsolete; computational constraints demand single-pass processing with bounded memory; and the requirement for timely predictions precludes the luxury of storing data for retrospective analysis [7]. These challenges have motivated extensive research into adaptive learning algorithms that can detect distribution changes and update their models accordingly.

While substantial progress has been made in developing adaptive classifiers, the dominant approaches rely on ensemble methods that achieve high accuracy at the cost of interpretability [21]. Methods such as Adaptive Random Forest (ARF) [21] and Streaming Random Patches (SRP) [22] combine hundreds of base learners through mechanisms that obscure the reasoning process. This opacity is problematic in high-stakes domains where understanding *why* a model makes specific predictions is as important as the predictions themselves [43]. The data streaming context introduces an additional dimension: the need to understand not only *what* a model predicts, but *how* and *why* its decision logic changes over time. For black-box methods, this adaptation process is entirely opaque. An interpretable classifier that maintains explicit decision rules offers transparency both in individual predictions and in the adaptation process, enabling practitioners to verify that model changes align with domain knowledge, identify genuine concept changes versus noise responses, and maintain an auditable history of decision logic evolution.

This paper introduces EGIS (Evolutionary Grammar for Interpretable Streams), an approach that addresses both interpretability requirements in data stream classification. EGIS leverages grammatical evolution [41] to maintain a population of rule-based classifiers expressed as human-readable IF-THEN rules, enabling domain experts to understand model decisions without specialized machine learning knowledge. The evolutionary process adapts these rules as new data arrives, with a multi-level self-adaptation framework that automatically adjusts parameters based on population diversity, performance history, and drift severity. A specialized mutation mechanism called Gene Therapy accelerates drift recovery by injecting discriminative patterns extracted from auxiliary decision trees. EGIS also introduces transition metrics that quantify how rules evolve between consecutive time periods, providing quantitative insight into the adaptation process.

The research questions addressed in this paper are formulated as follows:

RQ1: Can a grammatical evolution approach achieve competitive predictive performance on data streams with concept drift

L. M. Almeida is with the Centro de Informática, Universidade Federal de Pernambuco, Recife, Brazil (e-mail: lma3@cin.ufpe.br).

L. L. Minku is with the School of Computer Science, University of Birmingham, Birmingham, B15 2TT, UK.

while maintaining complete interpretability?

RQ2: How can the adaptation behavior of a rule-based classifier be quantified and analyzed to understand not only what changes occur but where, how, and when these changes manifest?

RQ3: Does a multi-level self-adaptation framework that responds to diversity, performance, and drift signals improve performance consistency across different types of concept drift?

The main contributions of this paper are as follows:

1) Interpretable Rule Evolution Framework: We propose a grammatical evolution approach that generates explicit IF-THEN classification rules, enabling domain experts to understand model decisions and verify classifier behavior without specialized machine learning knowledge.
2) Transition Metrics: We introduce three metrics, namely Transition Change Score (TCS), Rule Instability Rate (RIR), and Average Modification Severity (AMS), that quantify rule evolution dynamics, answering where, how, and when structural changes occur during concept drift adaptation.
3) Multi-Level Self-Adaptation Framework: We develop an adaptation system with fifteen mechanisms that respond to population diversity, performance history, and drift severity signals, enabling automatic parameter adjustment without manual tuning.
4) Gene Therapy Mutation: We propose a knowledge-guided mutation operator that extracts discriminative patterns from specialized decision trees and injects them directly into the evolutionary population, accelerating convergence during drift recovery.
5) Empirical Evaluation: Experiments spanning 48 datasets, five drift categories, seven EGIS configurations, and seven comparative methods, with Friedman tests and pairwise Wilcoxon signed-rank tests with Bonferroni correction.

The paper is organized as follows: Section II formalizes the problem; Section III reviews related work; Section IV describes EGIS; Sections V–VI present experiments and results; Section VII concludes.

II. Problem Formulation

We formalize the problem of interpretable data stream classification with concept drift adaptation, establishing the mathematical framework that underlies the proposed approach. Let $\mathcal{S} = \{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \ldots\}$ denote an infinite data stream where each instance $(\mathbf{x}_t, y_t)$ arrives at discrete time step t, with $\mathbf{x}_t \in \mathcal{X} \subseteq \mathbb{R}^d$ representing a d-dimensional feature vector and $y_t \in \mathcal{Y} = \{c_1, c_2, \ldots, c_K\}$ denoting the class label from a finite set of K classes. The stream is generated according to a time-varying joint probability distribution $P_t(\mathbf{x}, y)$ that may change at unknown time points, a phenomenon known as concept drift.

Concept drift occurs when $\exists t_1, t_2 : P_{t_1}(\mathbf{x}, y) \neq P_{t_2}(\mathbf{x}, y)$, manifesting as changes in the posterior $P(y|\mathbf{x})$ (real drift) or the marginal $P(\mathbf{x})$ (virtual drift) [20]. Drift dynamics range from abrupt (instantaneous) to gradual (extended transitions), with recurring concepts as a special case.

The stream is processed in discrete chunks $\mathcal{D}_i = \{(\mathbf{x}_t, y_t) : t \in [(i-1) \cdot n + 1, i \cdot n]\}$ of size n. The classifier $f : \mathcal{X} \to \mathcal{Y}$ is trained on chunk $\mathcal{D}_i$ and evaluated on $\mathcal{D}_{i+1}$ (*train-then-test* protocol), providing a realistic assessment of generalization to future data.

The interpretability requirement constrains the hypothesis space to rule-based classifiers of the form $\mathcal{R} = \{r_1, r_2, \ldots, r_m\}$, where each rule r_k takes the form:

$$r_k : \text{IF } \phi_k(\mathbf{x}) \text{ THEN } c_k \tag{1}$$

Here $\phi_k : \mathcal{X} \to \{0, 1\}$ is a Boolean function composed of conjunctions and disjunctions of atomic conditions, and $c_k \in \mathcal{Y}$ is the consequent class. Each atomic condition compares an attribute value against a threshold using a relational operator:

$$\text{attr}_j \circ v, \quad \circ \in \{<, >, \leq, \geq, =, \neq\} \tag{2}$$

This representation ensures that any evolved classifier can be directly inspected and understood by domain experts, as each rule expresses its decision logic in explicit terms that require no machine learning expertise to interpret.

The transition monitoring problem extends classification to analyze how rule sets evolve. Given consecutive rule sets $\mathcal{R}_{t-1}$ and $\mathcal{R}_t$, we quantify: (1) overall structural change magnitude, (2) the proportion of added, modified, or deleted rules, and (3) individual rule modification severity.

III. Related Work

This section reviews the main research areas related to EGIS: concept drift detection and adaptation, ensemble methods for data streams, rule-based and interpretable stream classifiers, evolutionary computation for classification, and explainability in dynamic environments.

A. Concept Drift Detection and Adaptation

Concept drift, defined as changes in the data-generating distribution over time, is the central challenge in data stream classification [20], [27], [35]. Detection methods can be broadly divided into two categories: explicit detectors that monitor error rates or distributional statistics, and passive approaches that continuously adapt without explicit detection signals [13].

Among explicit detectors, DDM [19] monitors the online error rate and signals drift when it exceeds a statistical threshold. ADWIN [5] maintains a variable-length sliding window and detects distributional change through subwindow comparison. Barros and Santos [3] present a large-scale comparison of drift detectors, highlighting the sensitivity-specificity trade-off: aggressive detectors react quickly but produce false alarms, while conservative detectors miss gradual changes. A comparative study by Gonçalves et al. [23] evaluates detectors across drift types, finding that no single method dominates all scenarios. Bayram et al. [4] provide a recent taxonomy of performance-aware detectors that extends this comparison to model degradation scenarios. Oliveira et al. [40] propose an adaptive Gaussian mixture model that distinguishes between virtual and real drifts, while type-driven detectors such as Type-LDD [48] attempt to classify the nature of detected changes.

Passive approaches sidestep this trade-off by continuously adjusting model parameters. Sliding window methods [33], weighted ensemble strategies [29], and diversity-driven approaches [38] adapt implicitly through mechanisms such as weight decay or ensemble pruning. However, most passive methods provide limited insight into *how much* or *what kind* of drift occurred. EGIS follows a passive strategy but extends it by quantifying drift severity through distributional similarity measures, enabling graduated responses that range from subtle parameter adjustments to partial population reinitialization.

B. Ensemble Methods for Data Streams

Ensemble methods have established the performance frontier in data stream classification [31]. Online Bagging and Boosting [42] extended batch ensemble techniques to the incremental setting by simulating bootstrap sampling through Poisson-distributed instance weights. Dynamic Weighted Majority (DWM) [29] adds and removes base learners based on their performance, weighting predictions by recent accuracy.

Adaptive Random Forest (ARF) [21] combines online bagging with Hoeffding trees equipped with ADWIN drift detectors on each base learner, enabling per-tree adaptation. Streaming Random Patches (SRP) [22] generalizes random subspaces and random patches to the streaming setting, achieving competitive performance through feature and instance subsampling. ROSE [8] targets imbalanced streams with adaptive online oversampling within the ensemble framework. CDCMS.CIL [9] uses model-space clustering to maintain ensemble diversity under non-stationary class distributions. Guo et al. [26] propose dynamical targeted ensemble learning that selects and combines base learners according to local competence regions. Learn^{++}.NSE [16] trains a new classifier on each data chunk and combines predictions using error-based weighting, handling both gradual and abrupt drift. These methods are typically implemented in frameworks such as River [39], which provides a unified Python API for streaming machine learning. Recent surveys have further explored ensemble learning for imbalanced streams [1] and the intersection with continual learning [25].

While these methods achieve strong predictive performance, their ensemble nature, which aggregates tens to hundreds of base learners, renders individual predictions unexplainable. This opacity is problematic in domains where understanding model decisions is a regulatory or operational requirement [43].

C. Rule-Based and Interpretable Stream Classifiers

Rule-based classifiers offer intrinsic interpretability by expressing decisions as IF-THEN statements. The Very Fast Decision Tree (VFDT) [14] introduced Hoeffding-bound-based incremental learning, later extended to rule learning by Very Fast Decision Rules (VFDR) [30]. The Hoeffding Adaptive Tree (HAT) [6] integrates ADWIN change detectors into Hoeffding tree nodes, replacing subtrees when drift is detected.

G-eRules [46] applies grammatical evolution to stream classification, demonstrating that evolutionary approaches can produce competitive rule-based classifiers. ERulesD2S [10] extends this line of work by incorporating concept drift detection and rule adaptation through genetic programming. However, ERulesD2S relies on a fixed parameterization without adaptive control, lacks mechanisms for knowledge injection during drift recovery, and provides no quantitative analysis of how rules evolve over time. ACDWM [34] maintains a weighted ensemble of chunk-based classifiers but is limited to binary classification tasks.

Recent work on cognitive biases in rule interpretation [28] underscores the importance of producing compact, well-structured rule sets that align with human reasoning limits, as overly complex or poorly organized rules can lead to systematic misinterpretation by domain experts.

A gap in the literature is the absence of rule-based stream classifiers that combine multi-level self-adaptation, knowledge-guided operators, and transition monitoring. Existing interpretable methods either lack drift adaptation entirely or employ simple fixed responses, whereas black-box ensembles adapt effectively but sacrifice transparency.

D. Evolutionary Computation for Classification

Grammatical evolution [41] constrains genetic programming through a context-free grammar that maps integer genotypes to syntactically valid programs. The separation of genotype and phenotype enables standard genetic operators while the grammar mapping produces well-formed structures [45]. Genetic programming has been applied to classification tasks across diverse domains [17], with rule-based representations offering a natural fit for interpretable models. Telikani et al. [47] provide a broader perspective on evolutionary machine learning across nine sub-fields, situating GP-based classification within the larger landscape. Mei et al. [36] survey the intersection of genetic programming and explainable AI, highlighting GP's capacity to produce interpretable symbolic models as a distinguishing advantage over black-box alternatives.

Multi-objective evolutionary algorithms such as NSGA-II [11] have been used to balance accuracy against complexity in classifier design. Parameter control in evolutionary algorithms, i.e., the automatic adjustment of mutation rates, selection pressure, and other operator parameters during the search, has been recognized as essential for robust performance across problem instances [15]. Approaches range from deterministic schedules to self-adaptive mechanisms where strategy parameters co-evolve with solution parameters. Zhou et al. [49] present a roadmap for evolutionary computation in explainable AI, identifying rule-based classifiers evolved by GP as a promising direction for intrinsically interpretable models.

EGIS builds on these foundations by combining grammatical evolution with a multi-level self-adaptation framework that adjusts fifteen parameters in response to population diversity, performance history, and drift severity. The Gene Therapy operator extends knowledge injection beyond random initialization by extracting discriminative patterns from auxiliary decision trees, a form of memetic computation [17] adapted to the streaming context.

E. Explainability in Dynamic Environments

The case for intrinsic interpretability over post-hoc explanations has been articulated by Rudin [43], who argues that inherently interpretable models should be preferred in high-stakes settings. Rudin et al. [44] formalize ten grand challenges for interpretable ML, reinforcing the argument for inherently transparent models and identifying the gap between post-hoc explanation quality and intrinsic model transparency. Surveys by Barredo Arrieta et al. [2] and Guidotti et al. [24] provide taxonomies of explainability methods, distinguishing model-agnostic post-hoc techniques (LIME, SHAP) from intrinsically interpretable architectures (decision trees, rule sets, linear models). Li et al. [32] complement these taxonomies from a data engineering perspective, distinguishing data-driven from knowledge-aware XAI approaches.

In the streaming context, explainability acquires a temporal dimension: practitioners need to understand not only *what* a model predicts at a given time, but *how* and *why* its decision logic changes across time periods. Freitas [18] discusses comprehensibility measures for classification models, noting that rule count and condition count are the primary proxies for human understanding. However, no prior work has proposed metrics to quantify the *dynamics* of rule evolution, namely the rate at which rules are replaced, the severity of modifications, and the overall adaptation intensity.

EGIS addresses this gap through three transition metrics (TCS, RIR, AMS) that capture complementary aspects of rule evolution between consecutive time periods. These metrics provide a quantitative framework for monitoring classifier adaptation, enabling practitioners to distinguish, for example, between a complete rule replacement strategy and a gradual refinement strategy, and to correlate adaptation intensity with drift events.

IV. Proposed Method: EGIS

EGIS employs grammatical evolution to maintain a population of rule-based classifiers that adapt to concept drift through the interplay of selection, crossover, mutation, and self-adaptation mechanisms. The complete processing pipeline is summarized in Algorithm 1, followed by detailed textual descriptions of each phase.

A. Grammar-Based Rule Representation

EGIS employs a context-free grammar $G = (V_N, V_T, P, S)$ to generate inherently interpretable classification rules. In BNF:

$$\begin{aligned}
\langle\text{ruleset}\rangle &::= \langle\text{rule}\rangle \mid \langle\text{rule}\rangle\langle\text{ruleset}\rangle \\
\langle\text{rule}\rangle &::= \text{IF } \langle\text{cond}\rangle \text{ THEN } \langle\text{class}\rangle \\
\langle\text{cond}\rangle &::= \langle\text{term}\rangle \mid \langle\text{term}\rangle\langle\text{logop}\rangle\langle\text{cond}\rangle \\
\langle\text{term}\rangle &::= \langle\text{attr}\rangle\langle\text{compop}\rangle\langle\text{value}\rangle \\
\langle\text{logop}\rangle &::= \text{AND} \mid \text{OR} \\
\langle\text{compop}\rangle &::= < \mid > \mid \leq \mid \geq \mid = \mid \neq
\end{aligned} \tag{3}$$

Each individual represents a complete rule set $\mathcal{R} = \{r_1, \ldots, r_m\}$ encoded as Abstract Syntax Trees (ASTs), where internal nodes contain logical operators and leaf nodes contain atomic conditions $\text{attr}_j \circ v$. This tree-based representation enables natural genetic operations while maintaining syntactic validity. Commutative operators are lexicographically sorted for canonical comparison. The grammar adapts to the feature space: numeric attributes use comparison operators with threshold values from the observed range; categorical attributes use equality operators. A bias toward conjunctive rules (80% AND vs 20% OR) produces more interpretable structures.

B. Multi-Objective Fitness Function

The fitness function balances three fundamental objectives that govern rule quality, with dynamic weighting that adapts based on drift context. The primary objective captures predictive accuracy through the geometric mean of class-specific recalls, a metric that ensures balanced performance across classes:

$$\text{G-Mean} = \left(\prod_{k=1}^{K} \text{Recall}_k \right)^{1/K} \tag{4}$$

where $\text{Recall}_k = \text{TP}_k / (\text{TP}_k + \text{FN}_k)$ for class k. The geometric mean penalizes classifiers that achieve high accuracy on majority classes while neglecting minority classes, an important consideration for imbalanced data streams.

The coverage objective incentivizes rules that classify a larger proportion of instances, preventing overly specific rules that match few examples:

$$\text{Coverage}(\mathcal{R}) = \frac{|\{(\mathbf{x}, y) \in \mathcal{D} : \exists r \in \mathcal{R}, \phi_r(\mathbf{x}) = 1\}|}{|\mathcal{D}|} \tag{5}$$

A class-weighted coverage bonus further rewards rules that cover underrepresented classes, with weights inversely proportional to class frequencies.

The complexity penalization objective favors simpler, more interpretable structures:

$$\text{Complexity}(\mathcal{R}) = \lambda_r \cdot |\mathcal{R}| + \lambda_c \cdot \sum_{r \in \mathcal{R}} |\phi(r)| + \lambda_f \cdot |A(\mathcal{R})| \tag{6}$$

where $|\mathcal{R}|$ counts total rules, $|\phi(r)|$ counts conditions in rule r, and $|A(\mathcal{R})|$ counts distinct attributes used across all rules. The coefficients λ_r, λ_c, and λ_f control the relative importance of each complexity component.

The complete fitness function combines these objectives with stability penalties that discourage unnecessary structural changes:

$$F(\mathcal{R}) = \alpha \cdot \text{G-Mean} + \beta_c \cdot \text{Cov} - \gamma \cdot \text{Cplx} - \beta_s \cdot \text{Stab} \tag{7}$$

where Cov, Cplx, and Stab denote Coverage, Complexity, and Stability respectively. The stability penalty measures feature distance from a reference configuration using the Jaccard distance:

$$\text{Stability}(\mathcal{R}) = d_J(A(\mathcal{R}), A_{\text{ref}}) = \frac{|A(\mathcal{R}) \triangle A_{\text{ref}}|}{|A(\mathcal{R}) \cup A_{\text{ref}}|} \tag{8}$$

where A_{ref} denotes the features used by the best individual from the previous generation.

The coefficients governing these objectives are not static but adapt dynamically based on drift severity. When significant concept change is detected, stability constraints are

relaxed to enable exploration of new rule structures. This adaptive weighting represents a key mechanism for balancing exploitation of learned knowledge with exploration of emerging patterns. Specifically, when drift severity exceeds a threshold, the stability penalty coefficient β_s is reduced to zero, and the complexity penalty is suspended, allowing the evolutionary process to discover radically different rule structures if needed.

An early termination mechanism interrupts evaluation of inferior individuals: if partial G-Mean (first 20% of instances) falls below 50% of elite median fitness, evaluation stops immediately.

C. Evolutionary Operators

The evolutionary process relies on carefully designed operators that balance exploration of the search space with exploitation of discovered solutions. Selection operates through a tournament mechanism with dynamically adjusted pressure. When population diversity is high, indicating risk of premature convergence, stronger selection pressure intensifies competition among individuals. When diversity diminishes, reduced pressure preserves alternative solutions that may prove valuable if conditions change. The tournament size varies according to:

$$k_t = k_{\min} + \delta \cdot (k_{\max} - k_{\min}) \tag{9}$$

where δ measures population diversity as the normalized fitness dispersion, and $k_{\min} = 2$, $k_{\max} = 5$ define the range.

The crossover operator uses a two-mode strategy: *expansion mode* (early generations, incomplete class coverage) performs subtree exchanges for broad exploration; *refinement mode* (later generations) transfers the best-performing rule from one parent to replace the worst in the child for each class.

Mutation operates at three structural levels within rules. Operator mutation alters comparison operators with probability p_{op}, enabling fine-grained adjustments to decision boundaries, for example, changing "$\text{attr}_j > v$" to "$\text{attr}_j \geq v$". Value mutation perturbs thresholds within valid ranges, allowing smooth adaptation to shifting data distributions. A mutation of magnitude ϵ drawn from a Gaussian distribution centered at zero adjusts the threshold while respecting attribute bounds. Subtree mutation replaces entire rule components with newly generated structures, introducing new structures when incremental adjustments prove insufficient. The overall mutation rate adapts inversely to population diversity:

$$p_m = p_m^{\text{base}} + (1 - \delta) \cdot (p_m^{\max} - p_m^{\text{base}}) \tag{10}$$

ensuring that mutation intensity increases as the population converges.

A specialized mechanism, Gene Therapy, trains a shallow decision tree per class and extracts discriminative rules scored by:

$$\text{Score}(r) = \text{Purity}(r) \cdot \log(1 + \text{Support}(r)) \tag{11}$$

The highest-scoring rule replaces the worst-performing rule for the corresponding class, injecting data-driven knowledge into the evolutionary population. This is particularly valuable during drift recovery.

Hill Climbing applies local search to elite individuals with intensity proportional to improvement potential: aggressive search for poor performers (G-Mean $< 85\%$), moderate for mid-range, and fine-tuning for high performers ($> 96\%$).

D. Self-Adaptation Framework

EGIS implements a multi-level self-adaptation system responding to three signals: population diversity δ (normalized fitness dispersion), performance history (three-level classification: good/medium/poor), and drift severity. When diversity diminishes, mutation rate increases and selection pressure decreases. Performance levels modulate the balance between knowledge inheritance and exploration.

Drift severity combines distributional similarity measures:

$$S_{\text{drift}} = 1 - (0.5 \cdot \text{sim}_\mu + 0.3 \cdot \text{sim}_\pi + 0.2 \cdot \text{sim}_\sigma) \tag{12}$$

where sim_μ, sim_π, sim_σ measure cosine/L1 similarity between feature means, class distributions, and feature standard deviations across chunks. Four severity levels trigger graduated responses: *stable* ($S < 0.05$, normal operation), *mild* (0.05–0.10, subtle adjustments), *moderate* (0.10–0.25, penalties removed, extra generations), and *severe* (≥ 0.25, partial reset with 60% reinitialization and 25 recovery generations).

The adaptive memory system maintains a best-solutions memory (with automatic abandonment on critical degradation) and a concept fingerprint memory for detecting recurring patterns. Population initialization uses a complexity-adaptive recipe based on a shallow decision tree probe, balancing seeded knowledge with random exploration.

A key design choice is implicit drift detection: rather than employing explicit detectors like ADWIN [5] or DDM [19], EGIS continuously adapts through fingerprint comparison and performance monitoring, avoiding the sensitivity-specificity trade-off inherent in threshold-based detection.

E. Transition Metrics

The transition metrics quantify rule evolution dynamics between consecutive chunks, characterizing the adaptation process itself. Rules from consecutive chunks $\mathcal{R}_{t-1}$ and $\mathcal{R}_t$ are matched using normalized Levenshtein similarity with threshold $\tau = 0.35$; above this threshold, rules are considered modifications of the same rule, and below it, complete replacements.

The Rule Instability Rate (RIR) measures the proportion of rules added or deleted between chunks:

$$\text{RIR} = \frac{|\text{Added}| + |\text{Deleted}|}{|\mathcal{R}_{t-1}| + |\mathcal{R}_t|} \tag{13}$$

High RIR indicates substantial structural turnover; low RIR indicates stability.

The Average Modification Severity (AMS) quantifies the degree of alteration for modified (not replaced) rules:

$$\text{AMS} = \frac{1}{|\text{Modified}|} \sum_{(r_{t-1}, r_t) \in \text{Modified}} (1 - \text{sim}(r_{t-1}, r_t)) \tag{14}$$

where similarity uses normalized Levenshtein edit distance: $\text{sim}(r_1, r_2) = 1 - \text{edit_distance}(\text{str}(r_1), \text{str}(r_2)) / \max(|\text{str}(r_1)|, |\text{str}(r_2)|)$.

The Transition Change Score (TCS) combines instability and severity into a unified adaptation intensity metric:

$$\text{TCS} = w_1 \cdot \text{RIR} + w_2 \cdot (1 - \text{RIR}) \cdot \text{AMS} \tag{15}$$

where $w_1 = 0.6$ and $w_2 = 0.4$ balance complete replacements and partial modifications. TCS near 1.0 indicates radical transformation; near 0.0 indicates minimal change. Joint interpretation enables rich analysis: high RIR with low AMS indicates a replacement strategy; low RIR with high AMS indicates gradual refinement. TCS peaks signal moments corresponding to concept drift events.

F. Algorithm Description

Algorithm 1 presents the complete EGIS processing pipeline. The algorithm processes the data stream in discrete chunks, updating the classifier as each chunk arrives.

The algorithm processes each chunk through eight phases. Phases 1–2 compute a statistical fingerprint of the current chunk, detect recurring concepts (similarity > 0.85 against stored fingerprints), and classify drift severity to control adaptation intensity. Phase 3 initializes the population using a complexity-adaptive recipe informed by a shallow decision tree probe, combining inherited knowledge with random exploration. Phase 4 executes the evolutionary loop with adaptive operators, Gene Therapy injection, and stagnation-triggered Hill Climbing. Phase 5 handles severe drift recovery through elevated mutation (0.5) and 60% reinitialization. Phases 6–8 extract the best classifier, compute transition metrics (Equations 13–15), and update memory structures.

Algorithm 1 EGIS: Evolutionary Grammar for Interpretable Streams

Require: Data stream $\mathcal{S}$, chunk size n, population size N
Ensure: Sequence of classifiers $\{f_1, f_2, \ldots\}$ and transition metrics

Initialize grammar G from feature space
Initialize empty memories: $\mathcal{M}_{\text{best}} \leftarrow \emptyset$, $\mathcal{M}_{\text{concept}} \leftarrow \emptyset$
Initialize previous rules $\mathcal{R}_{\text{prev}} \leftarrow \emptyset$
for each chunk $\mathcal{D}_i$ from stream $\mathcal{S}$ **do**
  // Phase 1: Fingerprint and Recurrence Detection
  $\text{fp}_i \leftarrow \text{ComputeFingerprint}(\mathcal{D}_i)$
  recurring $\leftarrow \text{DetectRecurrence}(\text{fp}_i, \mathcal{M}_{\text{concept}})$
  // Phase 2: Drift Severity Classification
  $S_{\text{drift}} \leftarrow \text{ComputeDriftSeverity}(\text{fp}_i, \text{fp}_{i-1})$
  level $\leftarrow \text{ClassifyDrift}(S_{\text{drift}})$
  Adjust penalties and parameters based on level
  // Phase 3: Population Initialization
  complexity $\leftarrow \text{ProbeComplexity}(\mathcal{D}_i)$
  $\mathcal{P}_0 \leftarrow \text{InitializePopulation}(N, \text{complexity}, \text{recurring}, \mathcal{M}_{\text{best}})$
  // Phase 4: Evolutionary Optimization
  for generation $g = 1$ to $g_{\max}$ **do**
    Evaluate fitness: $F(p) \leftarrow$ Equation 7 for all $p \in \mathcal{P}_{g-1}$
    $\delta \leftarrow \text{ComputeDiversity}(\mathcal{P}_{g-1})$
    Adjust tournament size and mutation rate based on δ
    $\mathcal{P}_{\text{elite}} \leftarrow \text{SelectElite}(\mathcal{P}_{g-1}, 0.1 \cdot N)$
    $\mathcal{P}_{\text{parents}} \leftarrow \text{TournamentSelection}(\mathcal{P}_{g-1})$
    $\mathcal{P}_{\text{offspring}} \leftarrow \text{AdaptiveCrossover}(\mathcal{P}_{\text{parents}}, \delta)$
    $\mathcal{P}_{\text{mutated}} \leftarrow \text{Mutation}(\mathcal{P}_{\text{offspring}}, p_m)$
    $\mathcal{P}_{\text{mutated}} \leftarrow \text{GeneTherapy}(\mathcal{P}_{\text{mutated}}, \mathcal{D}_i)$
    $\mathcal{P}_g \leftarrow \mathcal{P}_{\text{elite}} \cup \mathcal{P}_{\text{mutated}}$
    if stagnation detected **then**
      Apply Hill Climbing to elite individuals
    end if
  end for
  // Phase 5: Recovery (if severe drift)
  if level = SEVERE **then**
    Execute additional recovery generations with elevated mutation
  end if
  // Phase 6: Output and Memory Update
  $f_i \leftarrow \text{BestIndividual}(\mathcal{P}_{g_{\max}})$
  $\mathcal{R}_i \leftarrow \text{ExtractRules}(f_i)$
  // Phase 7: Transition Metrics
  Compute RIR, AMS, TCS using Equations 13-15
  $\mathcal{R}_{\text{prev}} \leftarrow \mathcal{R}_i$
  // Phase 8: Memory Management
  Update $\mathcal{M}_{\text{best}}$ with elite individuals
  Store $(\text{fp}_i, \text{elite})$ in $\mathcal{M}_{\text{concept}}$
  if abandonment criteria met **then**
    $\mathcal{M}_{\text{best}} \leftarrow \emptyset$
  end if
  Output: Classifier f_i, Rules $\mathcal{R}_i$, Metrics (RIR, AMS, TCS)
end for

V. Experimental Setup

This section describes the experimental setup for evaluating EGIS across 48 datasets spanning five drift categories.

A. Datasets

The experimental corpus comprises 48 data streams generated through the MOA framework [7] (synthetic) and real-world sources. Table I organizes datasets by drift category.

All synthetic datasets were generated once with fixed random seeds (12,000 instances each), partitioned into non-overlapping chunks of 500, 1,000, or 2,000 instances (24, 12, or 6 chunks respectively). Drifts occur at predetermined boundaries. The five categories span abrupt, gradual, noisy (10–20% label corruption), stationary, and real-world drift patterns.

B. Comparative Methods

We compare EGIS against seven baseline methods spanning ensemble approaches, tree-based methods, and interpretable classifiers:

- ROSE [8]: Robust Online Self-adjusting Ensemble for imbalanced streams, evaluated in both original and chunk-based evaluation configurations.
- ARF [21]: Adaptive Random Forest combining online bagging with adaptive Hoeffding trees.
- SRP [22]: Streaming Random Patches employing random subspaces.
- CDCMS.CIL [9]: Concept Drift handling based on Clustering in Model Space for Class-Imbalanced Learning, a heterogeneous ensemble that maintains diversity through model-space clustering with G-Mean weighted voting.
- HAT [6]: Hoeffding Adaptive Tree with ADWIN change detection.
- ACDWM [34]: Adaptive Chunk-based Dynamic Weighted Majority.
- ERulesD2S [10]: Evolutionary Rules for Data Streams, the only other interpretable baseline.

a) Implementation details: HAT, ARF, and SRP use River 0.21+ [39] with default parameters (ARF and SRP:

TABLE I
DATASET CATEGORIES AND CHARACTERISTICS

Category	Datasets
Abrupt Drift	SEA_Abrupt_Simple, SEA_Abrupt_Chain, SEA_Abrupt_Recurring, AGRAWAL_Abrupt_Simple_Mild, AGRAWAL_Abrupt_Simple_Severe, AGRAWAL_Abrupt_Chain_Long, RBF_Abrupt_Blip, RBF_Abrupt_Severe, HYPERPLANE_Abrupt_Simple, STAGGER_Abrupt_Chain, STAGGER_Abrupt_Recurring, RANDOMTREE_Abrupt_Simple, RANDOMTREE_Abrupt_Recurring, SINE_Abrupt_Simple, LED_Abrupt_Simple, WAVEFORM_Abrupt_Simple
Gradual Drift	SEA_Gradual_Simple_Fast, SEA_Gradual_Simple_Slow, SEA_Gradual_Recurring, RBF_Gradual_Moderate, RBF_Gradual_Severe, HYPERPLANE_Gradual_Simple, STAGGER_Gradual_Chain, RANDOMTREE_Gradual_Simple, SINE_Gradual_Recurring, LED_Gradual_Simple, WAVEFORM_Gradual_Simple
Noisy Drift	AGRAWAL_Abrupt_Simple_Severe_Noise, RBF_Abrupt_Blip_Noise, RBF_Gradual_Severe_Noise, HYPERPLANE_Gradual_Noise, RANDOMTREE_Gradual_Noise, SEA_Abrupt_Chain_Noise, STAGGER_Abrupt_Chain_Noise, SINE_Abrupt_Recurring_Noise
Stationary	AGRAWAL_Stationary, SEA_Stationary, RBF_Stationary, HYPERPLANE_Stationary, STAGGER_Stationary, RANDOMTREE_Stationary, SINE_Stationary, LED_Stationary, WAVEFORM_Stationary
Real-World	Electricity, AssetNegotiation_F2, AssetNegotiation_F3, AssetNegotiation_F4

`n_models=10`). ROSE executes via the MOA Java CLI with default settings. ACDWM uses the DWMIL library (θ=0.001, err=G-Mean, binary only). ERulesD2S uses MOA/JCLEC4 (population=25, generations=50, rules/class=5). CDCMS uses the CDCMS_CIL_GMean variant.

b) Model Limitations: ACDWM supports only binary classification; CDCMS also failed on multiclass datasets. Following [12], we assign G-Mean=0.0 to failed cases. ERulesD2S frequently produced collapsed classifiers. Ensemble methods provide no interpretability, representing the trade-off motivating EGIS.

C. Experimental Configurations

Table II presents the seven EGIS configurations varying chunk size and complexity penalty.

Smaller chunks (500) enable more frequent adaptation; larger chunks (2000) provide more stable training. The complexity penalty ($\gamma \in \{0.0, 0.1, 0.3\}$) controls the trade-off between accuracy and rule simplicity, with $\gamma = 0.3$ imposing the strongest pressure toward compact rule sets.

TABLE II
EXPERIMENTAL CONFIGURATIONS

Config	Chunk	Pen.	Sets	Focus
EXP-500-NP	500	0.0	48	Fine-grained
EXP-500-P	500	0.1	48	Interpretability
EXP-500-P03	500	0.3	48	Strong penalty
EXP-1000-NP	1000	0.0	48	Balanced
EXP-1000-P	1000	0.1	48	Complexity ctrl
EXP-2000-NP	2000	0.0	48	Larger windows
EXP-2000-P	2000	0.1	48	Larger + penalty

EGIS uses a population of 120 individuals evolved for up to 200 generations per chunk, with 10% elitism, adaptive tournament size (2–5), and base mutation rate 0.1. Maximum 15 rules per class with tree depth ≤ 10 ensures interpretability. Complete hyperparameter settings for EGIS and all comparative methods are provided in the Supplementary Material.

D. Evaluation Protocol

EGIS employs a train-then-test protocol: for each chunk $\mathcal{D}_i$, the evolutionary algorithm produces a rule-based classifier evaluated on the subsequent chunk $\mathcal{D}_{i+1}$. Comparative streaming methods (ARF, SRP, HAT, ROSE, CDCMS.CIL) operate in their native prequential mode with instance-by-instance updates. Each method is evaluated according to its designed operational paradigm, ensuring fair comparison by leveraging each algorithm's native learning mechanism. Both paradigms exploit temporal continuity: EGIS through explicit elite solution inheritance across chunks, and prequential methods through persistent model state.

E. Evaluation Metrics

The primary metric is G-Mean (Equation 4), balancing sensitivity and specificity across classes. Statistical significance is assessed through the Friedman test followed by pairwise Wilcoxon signed-rank tests with Bonferroni correction [12]. We also conduct stratified analysis by drift type (abrupt, gradual, noisy, stationary, real) to identify method-specific strengths.

VI. RESULTS AND DISCUSSION

This section presents experimental results spanning 48 datasets across five drift categories, with statistical validation. We analyze EGIS performance across multiple configurations, compare against seven baseline methods, and examine the explainability characteristics of the proposed approach.

A. Overall Performance Comparison

Table III presents the summary performance comparison across experimental configurations. On the 41 binary datasets where all models can be compared fairly, EGIS achieves G-Mean of 0.859 (EXP-500), 0.869 (EXP-1000), and 0.855 (EXP-2000), competitive with leading ensemble methods while providing complete interpretability. EGIS substantially outperforms ERulesD2S, the only other interpretable baseline, by over 25 percentage points across all chunk sizes.

TABLE III
SUMMARY PERFORMANCE ACROSS ALL EXPERIMENTS (G-MEAN). TOP SECTION: 41 BINARY DATASETS WHERE ALL 8 MODELS CAN BE EVALUATED. BOTTOM SECTION: ALL 48 DATASETS (INCLUDING 7 MULTICLASS) WITH ONLY THE 6 MODELS THAT SUPPORT MULTICLASS CLASSIFICATION (ACDWM AND CDCMS EXCLUDED AS THEY ARE BINARY-ONLY). BOLD INDICATES BEST INTERPRETABLE METHOD; UNDERLINE INDICATES BEST OVERALL.

	EXP-500		EXP-1000		EXP-2000	
Model	Mean	Std	Mean	Std	Mean	Std
Binary only (n=41):						
ROSE	0.892	0.111	0.892	0.111	0.892	0.111
ARF	0.878	0.141	0.879	0.139	0.878	0.140
SRP	0.869	0.150	0.862	0.152	0.862	0.153
ACDWM	0.860	0.093	0.818	0.078	0.740	0.069
CDCMS	0.846	0.152	0.841	0.152	0.839	0.153
HAT	0.815	0.150	0.819	0.146	0.820	0.148
EGIS	**0.859**	**0.125**	**0.869**	**0.113**	**0.855**	**0.108**
ERulesD2S	0.605	0.087	0.599	0.084	0.598	0.082
All datasets (n=48):						
ARF	0.862	0.140	0.873	0.131	0.870	0.133
EGIS	**0.856**	**0.124**	**0.866**	**0.114**	**0.855**	**0.110**
ROSE	0.855	0.170	0.855	0.170	0.855	0.170
SRP	0.846	0.165	0.848	0.152	0.850	0.150
HAT	0.767	0.231	0.770	0.230	0.772	0.229
ERulesD2S	0.562	0.137	0.556	0.136	0.556	0.134

A key finding is EGIS's improvement over ERulesD2S, the only other interpretable baseline, with a performance gap of over 25 percentage points (0.859 vs 0.605 on binary datasets). This demonstrates that the proposed self-adaptation framework and Gene Therapy mechanism yield markedly better rule evolution than prior evolutionary approaches. On the 41 binary datasets, EGIS achieves competitive G-Mean in EXP-500 (0.859), EXP-1000 (0.869), and EXP-2000 (0.855), addressing **RQ1** by demonstrating that interpretability does not require sacrificing predictive accuracy. When all 48 datasets are considered (including 7 multiclass, evaluated only on the 6 models that support multiclass), EGIS achieves the highest mean G-Mean among interpretable methods (0.856 in EXP-500, 0.866 in EXP-1000, 0.855 in EXP-2000), closely matching ensemble methods such as ARF (0.862, 0.873, 0.870).

The complete per-dataset G-Mean comparison across all 41 binary datasets for each model is provided in the Supplementary Material (Table S1). EGIS wins against all 41 datasets versus eRulesD2S and 21/17/3 (W/L/D) versus CDCMS. Against ensemble methods, EGIS shows 11/30/0 against ARF and 7/32/2 against ROSE; however, as shown in Table VIII, these differences are not statistically significant after Bonferroni correction (adjusted p=0.530 and p=0.0006, respectively; only ROSE reaches significance).

B. Performance by Drift Type

Table IV presents EGIS performance stratified by drift category. Analysis reveals consistent performance across different drift scenarios, with the highest performance on synthetic datasets with known drift patterns.

On the 41 binary datasets evaluated in Table IV, EGIS achieves G-Mean of 0.859 overall, with consistent performance across all drift categories. Real-world datasets achieve the highest performance (0.890), likely due to well-defined class boundaries in the AssetNegotiation and Electricity streams.

TABLE IV
PERFORMANCE BY DRIFT TYPE (G-MEAN, EXP-500 CONFIGURATION). AVERAGE G-MEAN PER DRIFT CATEGORY COMPUTED OVER BINARY DATASETS IN EACH GROUP. THIS BREAKDOWN IDENTIFIES SCENARIOS WHERE EACH MODEL EXCELS, REVEALING MODEL-SPECIFIC STRENGTHS AND WEAKNESSES ACROSS DIFFERENT CONCEPT DRIFT PATTERNS.

Type	**EGIS**	**ARF**	**HAT**	**SRP**	**ROSE**	**eRules**	**CDCMS**	**ACDWM**
Abrupt (14)	0.860	0.863	0.776	0.857	0.888	0.600	0.813	0.861
Gradual (9)	0.855	0.888	0.832	0.875	0.900	0.607	0.859	0.859
Noisy (8)	0.846	0.855	0.793	0.847	0.888	0.593	0.818	0.849
Stationary (6)	0.862	0.883	0.835	0.877	0.901	0.602	0.889	0.860
Real (4)	0.890	0.941	0.927	0.929	0.885	0.649	0.926	0.882
Overall (41)	**0.859**	0.878	0.815	0.869	0.892	0.605	0.846	0.860

Stationary datasets (0.862) and abrupt drift (0.860) yield similar results, confirming consistent adaptation. The slightly lower noisy drift performance (0.846) reflects the additional challenge of learning from corrupted labels.

C. Rule Complexity Analysis

Table V presents interpretability metrics for EGIS across configurations, including the effect of the complexity penalty.

TABLE V
EGIS RULE COMPLEXITY BY CONFIGURATION

Config	**Avg Rules**	**Cond/Rule**	**AND**	**OR**
EXP-500-NP (γ=0.0)	15.0±5.1	5.30±1.85	68.4	3.6
EXP-500-P (γ=0.1)	13.5±4.9	4.62±1.52	54.4	1.7
EXP-500-P03 (γ=0.3)	12.1±4.2	4.65±1.68	51.3	2.5
EXP-1000-NP (γ=0.0)	19.7±8.0	5.67±1.99	98.6	3.3
EXP-1000-P (γ=0.1)	19.6±8.0	5.65±1.98	98.2	3.3
EXP-2000-NP (γ=0.0)	23.6±14.5	5.86±2.04	126.6	3.4
EXP-2000-P (γ=0.1)	17.9±8.9	5.34±1.88	92.5	2.8

EGIS produces compact rule sets with 12–24 rules on average, each containing approximately 5–6 conditions. This complexity level falls within human cognitive limits for comprehension [37], enabling domain experts to inspect and validate the complete classifier. The complexity penalty reduces the average number of conditions per rule from 5.30 to 4.62 (γ=0.1) and 4.65 (γ=0.3) in EXP-500, demonstrating its effectiveness in producing simpler rules. With γ=0.3, the average number of rules drops to 12.1 (from 15.0 at γ=0.0), a 19% reduction. Larger chunk sizes increase complexity: EXP-2000-NP produces 23.6 rules on average (vs 15.0 for EXP-500-NP), as more training data enables more detailed rule sets. The predominance of AND operators (51–127 per rule set) over OR operators (2–4) indicates that EGIS favors conjunctive rules, which are generally easier for humans to interpret.

1) Rule Interpretability Examples: To demonstrate the concrete interpretability of EGIS, Table VI presents representative rules extracted from actual experiments, organized by complexity level. These examples illustrate how domain experts can directly inspect and validate the learned decision logic.

These rules demonstrate EGIS's interpretability across complexity levels. Simple rules like "IF nswprice $>$ 0.0912 THEN Class 1" are immediately actionable. Even the most

TABLE VI
REPRESENTATIVE RULES EXTRACTED BY EGIS

Dataset	Rule	Cond.
Simple Rules (1-2 conditions)		
Electricity	IF nswprice $>$ 0.0912 THEN Class 1	1
AGRAWAL	IF salary $\leq$ 24833 THEN Class 0	1
Electricity	IF nswprice $>$ 0.1222 AND nswdemand $>$ 0.34 THEN Class 1	2
Medium Complexity Rules (3-5 conditions)		
Electricity	IF nswprice $>$ 0.0842 AND date $>$ 0.0049 AND nswprice $>$ 0.0905 THEN Class 1	3
AGRAWAL	IF salary $\leq$ 48849 AND age $>$ 59.5 AND salary $>$ 25040 THEN Class 1	3
AGRAWAL	IF salary $\in$ (48090, 119828] AND age $\leq$ 59.5 AND salary $\in$ (64834, 106310] THEN Class 1	5
High Complexity Rules (6+ conditions)		
SEA	IF attr1 $\leq$ 5.06 AND attr0 $\leq$ 5.44 AND attr0 $\leq$ 4.59 AND attr1 $\leq$ 4.42 AND attr0 $>$ 3.24 AND attr1 $>$ 3.57 THEN Class 1	6
AGRAWAL	IF salary $\in$ (48849, 119828] AND salary $\leq$ 100160 AND age $>$ 39.5 AND loan $>$ 412323 AND salary $\leq$ 81245 AND age $>$ 59.5 AND salary $>$ 73105 THEN Class 1	8

complex rules (8 conditions) remain fully inspectable, a capability impossible with ensemble approaches.

Figures 1 and 2 present detailed case studies for two representative datasets, illustrating how EGIS adapts its rule sets to different drift types.

The case studies in Figures 1–2 reveal complementary adaptation patterns. Panel (a) shows that EGIS maintains competitive accuracy throughout both scenarios, with brief dips at drift points followed by rapid recovery. In the gradual drift scenario (RBF_Gradual_Moderate), rule changes are distributed smoothly across the drift period, with moderate TCS and RIR values. In contrast, STAGGER_Abrupt_Chain exhibits sharp TCS and RIR spikes at drift points, indicating rapid rule replacement. Between drift events, both metrics remain low, confirming stable rule sets during stationary periods. These contrasting patterns demonstrate that EGIS automatically calibrates its adaptation intensity to the severity and type of concept drift.

D. Statistical Significance Analysis

The Friedman test on 41 binary datasets reveals significant differences among the eight methods ($\chi^2(7) = 123.1$, $p < 0.001$), with a Nemenyi critical difference of CD = 1.64. Figure 3 presents the critical difference diagram showing statistically significant groupings among methods.

Table VII presents the complete model ranking based on the Friedman test across 41 binary datasets (the primary comparison, as ACDWM and CDCMS are binary-only methods that would be unfairly penalized on multiclass datasets). ROSE achieves the best average rank (2.41), followed by ARF (2.85), SRP (3.68), and EGIS (4.45). EGIS ranks fourth overall and is the best interpretable method, with a rank gap to ARF (1.60) that falls within the critical distance (CD = 1.64), indicating no statistically significant difference. The ranking reveals clear performance tiers: ensemble methods (ROSE, ARF, SRP) form the top group, followed by a competitive middle tier comprising EGIS (4.45), CDCMS (4.65), and ACDWM (4.95), with HAT (5.49) and ERulesD2S (7.51) further behind.

TABLE VII
COMPLETE MODEL RANKING BASED ON FRIEDMAN TEST (BINARY DATASETS, N=41, EXP-500-NP)

Rank	Model	Avg Rank	G-Mean	Std	W	L	D
1	ROSE	2.41	0.892	0.111	7	32	2
2	ARF	2.85	0.878	0.141	11	30	0
3	SRP	3.68	0.869	0.150	13	28	0
4	**EGIS**	**4.45**	**0.859**	**0.125**	–	–	–
5	CDCMS	4.65	0.846	0.152	21	17	3
6	ACDWM	4.95	0.860	0.093	24	17	0
7	HAT	5.49	0.815	0.150	26	13	2
8	eRulesD2S	7.51	0.605	0.087	41	0	0

Friedman Test: $\chi^2(7) = 123.1$, $p < 0.001$
Critical Distance (Nemenyi, α=0.05): CD = 1.64

Table VIII presents pairwise Wilcoxon signed-rank test results with Bonferroni correction.

TABLE VIII
STATISTICAL SIGNIFICANCE TESTS (PAIRWISE WILCOXON WITH BONFERRONI, BINARY DATASETS, EXP-500-NP)

Comparison	p-value	Adj. p	Sig.	Mean Δ	Effect
EGIS vs ARF	0.0757	0.5298	No	-0.018	-0.17 (s)
EGIS vs SRP	0.0825	0.5773	No	-0.010	-0.12 (n)
EGIS vs HAT	0.0274	0.1921	No	+0.044	0.18 (s)
EGIS vs ROSE	¡0.0001	0.0006	Yes	-0.033	-0.18 (s)
EGIS vs ACDWM	0.9535	1.0000	No	-0.001	0.13 (n)
EGIS vs ERulesD2S	¡0.0001	¡0.0001	Yes	+0.254	0.84 (l)
EGIS vs CDCMS	0.5908	1.0000	No	+0.013	0.03 (n)

Pairwise Wilcoxon tests with Bonferroni correction (Table VIII) reveal that only two comparisons are statistically significant: EGIS vs ERulesD2S (Cliff's $\delta = 0.84$, large effect, adjusted p < 0.0001) and EGIS vs ROSE ($\delta = -0.18$, small effect, adjusted p = 0.0006). The comparison with ERulesD2S confirms that EGIS advances interpretable stream classification. Against the top ensemble methods ARF ($\delta = -0.17$, small, adjusted p = 0.530) and SRP ($\delta = -0.12$, negligible, adjusted p = 0.577), the differences are *not* statistically significant after Bonferroni correction, indicating that EGIS achieves statistical parity with these black-box methods while providing complete interpretability. The comparisons with HAT ($\delta = 0.18$, small), ACDWM ($\delta = 0.13$, negligible), and CDCMS ($\delta = 0.03$, negligible) are also not significant.

E. Transition Metrics Analysis

Table IX presents the transition metrics by drift type, quantifying how EGIS's adaptation behavior varies with drift characteristics.

The transition metrics reveal meaningful patterns in adaptation behavior. Abrupt drift triggers the highest transition activity (TCS = 0.248, RIR = 0.273), reflecting the need for substantial rule replacement during sudden distribution changes. Noisy drift shows similarly elevated values (TCS = 0.246, RIR = 0.255), as noise-corrupted boundaries require frequent recalibration. Gradual and stationary scenarios yield lower values (TCS = 0.228–0.229, RIR = 0.233–0.248), reflecting

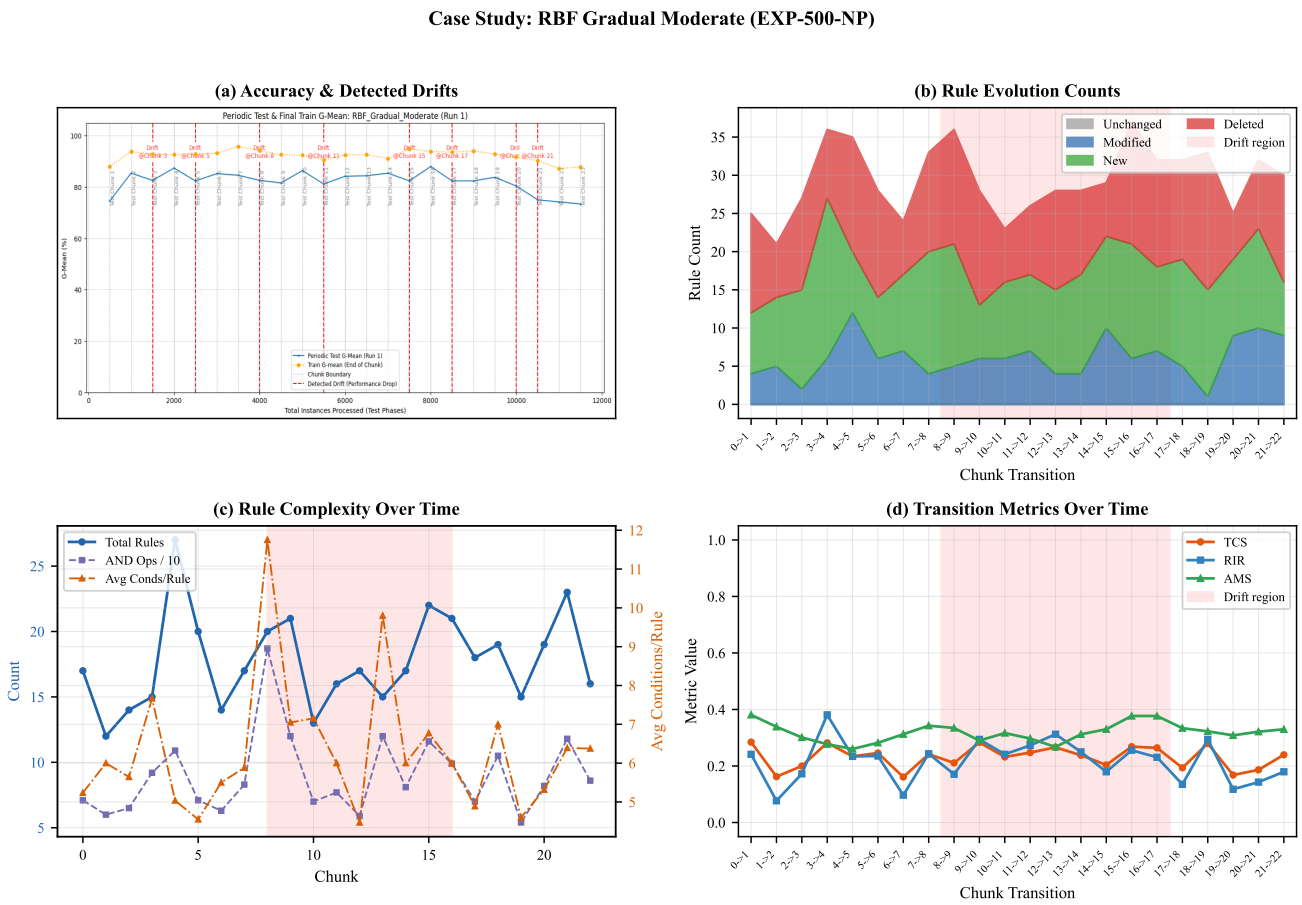

Fig. 1. Case study: RBF_Gradual_Moderate (EXP-500-NP). (a) Per-chunk test accuracy with detected drifts marked. (b) Rule evolution counts showing unchanged, modified, new, and deleted rules per transition. (c) Rule complexity over time (total rules, conditions per rule). (d) Transition metrics (TCS, RIR, AMS); the shaded region indicates the gradual drift period.

TABLE IX
TRANSITION METRICS BY DRIFT TYPE (EGIS)

Drift Type	TCS	RIR	AMS
Abrupt	0.248±0.115	0.273±0.202	0.284±0.096
Gradual	0.228±0.102	0.233±0.180	0.284±0.084
Noisy	0.246±0.099	0.255±0.184	0.301±0.066
Stationary	0.229±0.121	0.248±0.206	0.264±0.099
Real	0.216±0.083	0.216±0.148	0.265±0.059

smoother adaptation. Real-world datasets show the lowest TCS (0.216) and RIR (0.216), suggesting that real drift patterns, while complex, may involve less radical restructuring than synthetic abrupt changes.

The RIR values (0.22–0.27) indicate that approximately 22–27% of rules change between consecutive chunks, with higher turnover during abrupt and noisy drift scenarios. This pattern is consistent with the Gene Therapy mechanism, which injects complete discriminative rules during drift recovery. These findings address **RQ2**: the transition metrics successfully quantify rule evolution dynamics, revealing distinct adaptation signatures for different drift types.

Figure 4 shows the distribution of all three transition metrics across drift types. Abrupt drift exhibits higher RIR (more rule replacements), while gradual scenarios show relatively higher AMS (more modifications). Real-world datasets display the widest distributions, reflecting diverse underlying drift patterns.

The patterns observed in Figure 4 for EXP-500 are consistent across all configurations (see Supplementary Material for detailed evolution heatmaps and TCS time series across drift types and chunk sizes).

F. EGIS Configuration Analysis

Table X examines the effect of the complexity penalty on EGIS performance.

TABLE X
EGIS COMPLEXITY PENALTY EFFECT (G-MEAN, MEAN±STD, 48 DATASETS)

Chunk	γ=0.0	Penalty	γ	Δ	p-value
500	0.856±0.123	0.855±0.121	0.1	+0.001	0.115
500	0.856±0.123	0.836±0.137	0.3	+0.020	<0.001*
1000	0.866±0.112	0.866±0.112	0.1	−0.001	0.646
2000	0.855±0.109	0.847±0.113	0.1	+0.008	0.001*

*Statistically significant (Wilcoxon signed-rank test, α=0.05)

A mild complexity penalty (γ=0.1) has negligible effect on predictive performance for EXP-500 (Δ=+0.001, p=0.115) and

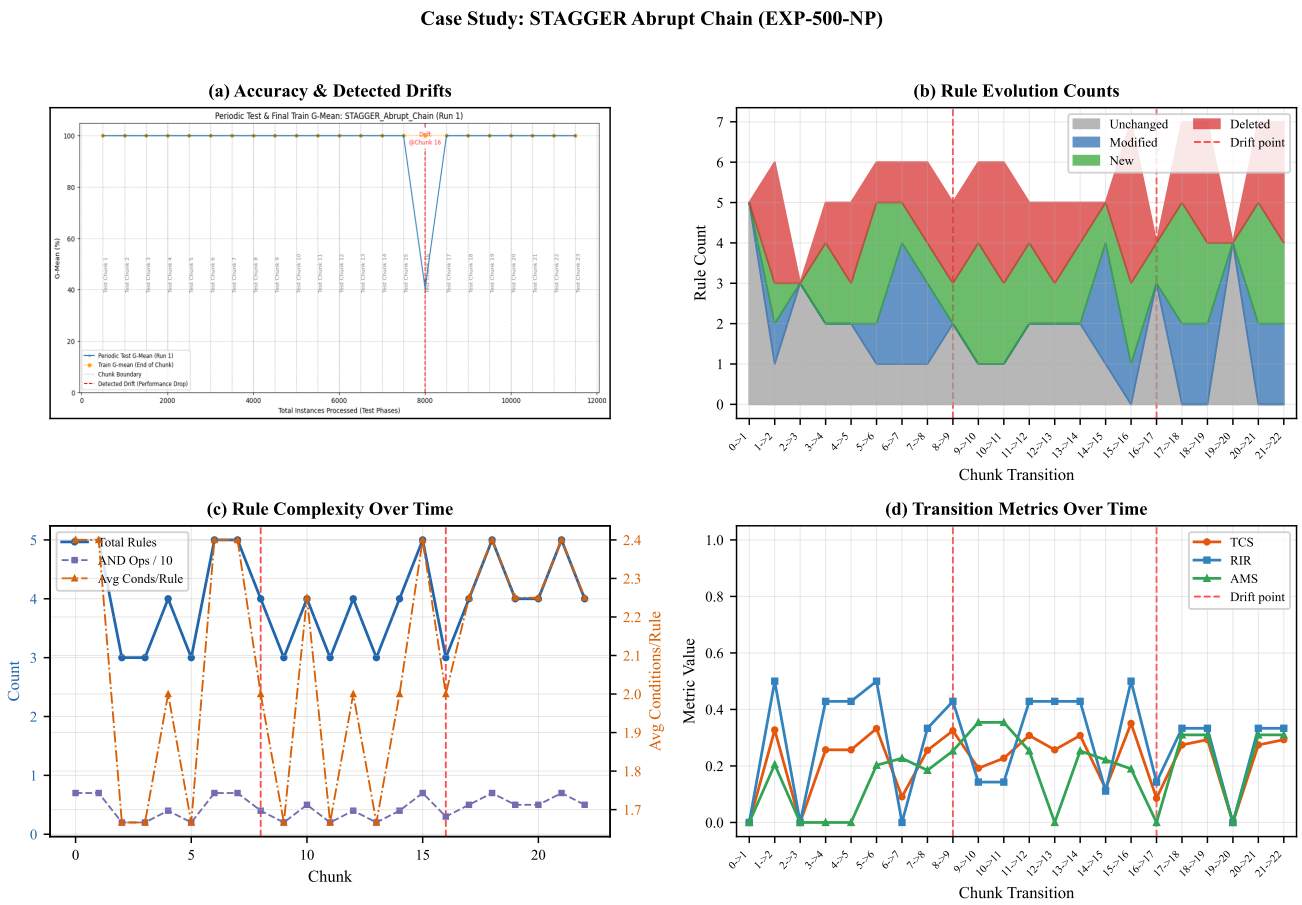

Fig. 2. Case study: STAGGER_Abrupt_Chain (EXP-500-NP). (a) Per-chunk test accuracy with detected drifts. (b) Rule evolution counts. (c) Rule complexity over time. (d) Transition metrics with drift points (dashed vertical lines) at chunks 8 and 16. TCS and RIR spike at drift points, confirming rapid rule replacement.

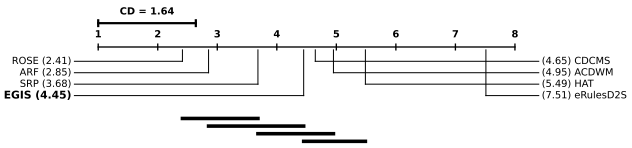

Fig. 3. Critical difference diagram on 41 binary datasets (Nemenyi, $\alpha = 0.05$). Connected methods are not significantly different. See Supplementary Material for multiclass analysis.

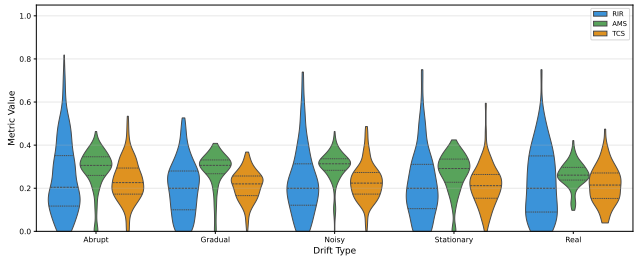

Fig. 4. Violin plots of RIR (blue), AMS (green), and TCS (orange) by drift type (EXP-500). RIR consistently dominates AMS, confirming adaptation primarily through rule replacement. Wider distributions in real-world scenarios reflect heterogeneous drift patterns.

EXP-1000 (Δ=$-$0.001, p=0.646), confirming that practitioners can enable the penalty to obtain simpler rules (4.62 vs 5.30 conditions per rule) without sacrificing accuracy. However, two penalty configurations show statistically significant effects: EXP-2000 with γ=0.1 (Δ=+0.008, p=0.001) and EXP-500 with γ=0.3 (Δ=+0.020, p$<$0.001). The strong penalty (γ=0.3) reduces G-Mean by 2 percentage points while producing the most compact rules (12.1 rules, 4.65 conditions), representing a meaningful interpretability-accuracy trade-off.

Sensitivity analysis across chunk sizes (500, 1000, 2000) and penalty settings confirms that EGIS performance is stable across configuration choices. EXP-1000 achieves the highest G-Mean (0.866) while EXP-500 (0.856) and EXP-2000 (0.855) perform similarly. Larger chunks (1000–2000) provide marginally higher G-Mean on stationary and gradual drift scenarios, while smaller chunks (500) enable faster adaptation to abrupt changes. These results address **RQ3**: the multi-level self-adaptation framework yields consistent performance across drift types and configuration choices. Complete configuration comparisons and per-dataset results for all chunk sizes are provided in the Supplementary Material.

G. Discussion: The Interpretability-Performance Trade-off

The results reveal a nuanced picture of the interpretability-performance trade-off in data stream classification. On 41 binary datasets, EGIS (rank 4.45) falls within the Nemenyi critical distance (CD=1.64) of ARF (2.85) and SRP (3.68), meaning these differences are *not* statistically significant. Only ROSE (2.41) achieves a significant advantage (adjusted p=0.0006). While EGIS loses more often than it wins against ROSE (7W/32L) and ARF (11W/30L), the performance gaps are small (Cliff's $\delta = -0.18$ and -0.17, both small effects), confirming competitive rather than dominant performance. This result addresses **RQ1**: interpretability does not require a statistically significant sacrifice in predictive accuracy compared to the leading ensemble methods.

At the same time, EGIS produces explicit IF-THEN rules that domain experts can directly inspect, validate, and audit. This is a capability that ensemble methods, which aggregate predictions from tens to hundreds of base learners, cannot provide. The improvement over ERulesD2S, the only other interpretable baseline, exceeds 25 percentage points (0.859 vs 0.605, Cliff's $\delta = 0.84$, large effect), stemming from the self-adaptation framework, Gene Therapy knowledge injection, and adaptive memory management.

Unlike black-box methods where adaptation is opaque, EGIS's transition metrics quantify how the classifier changes over time. Practitioners can observe that approximately 22–27% of rules change between consecutive chunks, with higher turnover during abrupt drift events (RIR = 0.273). These metrics enable monitoring of classifier evolution and correlation of adaptation intensity with drift events, addressing the temporal explainability dimension identified in the related work. These results position EGIS as the method of choice when explainability requirements preclude black-box alternatives. Extended per-dataset results and additional visualizations are provided in the Supplementary Material.

H. Visual Analysis

Figure 5 presents the performance distribution across datasets for each model. EGIS exhibits competitive median performance with ensemble methods while consistently outperforming ERulesD2S across all datasets.

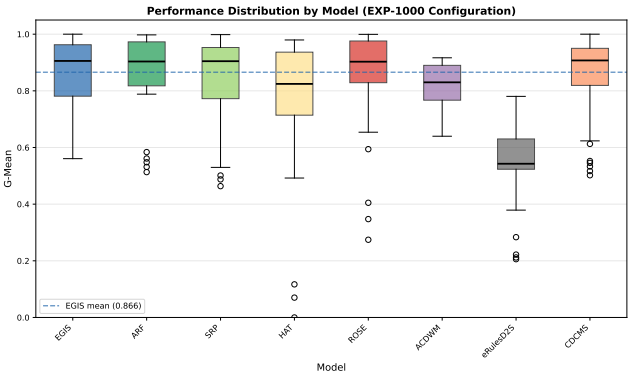

Fig. 5. Performance distribution by model (G-Mean, EXP-500). EGIS is evaluated across all 48 datasets; binary-only methods (ACDWM, CDCMS) on 41 datasets.

Per-chunk G-Mean evolution, visible in panel (a) of Figures 1–2, confirms rapid recovery after drift points: accuracy dips briefly at concept changes before stabilizing at competitive levels. Extended per-dataset temporal evolution plots are provided in the Supplementary Material.

VII. Conclusion

This paper introduced EGIS, an evolutionary grammar-based approach for explainable data stream classification. Through experiments across 48 datasets spanning five drift categories and comparisons against seven state-of-the-art methods, we addressed the research questions posed in the introduction.

Regarding RQ1, EGIS achieves competitive predictive performance (G-Mean of 0.859 on binary datasets in EXP-500) while providing complete interpretability through explicit IF-THEN rules. On 41 binary datasets, EGIS ranks fourth (average rank 4.45) and falls within the Nemenyi critical distance (CD = 1.64) of ARF (2.85), indicating no statistically significant difference from ARF or SRP. Only ROSE (2.41) achieves a significant advantage. EGIS outperforms ERulesD2S, the only other interpretable baseline, by over 25 percentage points (0.859 vs 0.605), demonstrating that the proposed mechanisms advance the state of the art in interpretable stream classification.

Regarding RQ2, the transition metrics (TCS, RIR, AMS) successfully quantify rule evolution dynamics. TCS captures composite adaptation intensity, with abrupt drift scenarios showing the highest values (TCS = 0.248) and real-world datasets the lowest (TCS = 0.216). RIR indicates that 22–27% of rules are replaced between consecutive chunks, with higher turnover during abrupt drift (RIR = 0.273). These metrics provide detailed quantitative insight into classifier adaptation behavior, enabling practitioners to understand not only what the model predicts but how its decision logic evolves over time.

Regarding RQ3, the multi-level self-adaptation framework demonstrates consistent performance across different drift types, with consistent behavior across abrupt, gradual, noisy, stationary, and real-world scenarios. The complexity penalty (γ=0.1) produces simpler rules (4.62 vs 5.30 conditions per rule) with negligible performance impact (<0.2%, not statistically significant for EXP-500 and EXP-1000), confirming that interpretability can be enhanced without sacrificing accuracy. A stronger penalty (γ=0.3) further reduces complexity at a modest accuracy cost (2 percentage points).

Future work will explore several directions. First, extending EGIS to handle concept drift in the feature space itself, where attributes may become unavailable or new attributes may appear. Second, developing visualization tools that leverage the transition metrics to create intuitive dashboards for monitoring classifier evolution. Third, investigating hybrid approaches that combine EGIS's interpretability with ensemble diversity mechanisms to close the remaining performance gap with black-box methods.

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